

Mini Assignment: Visual Breakdown

Human-Computer Interaction (HCI) vs. Computer Graphics (CG)

Selected Application: Google Earth

Format: 2-Slide Analysis + Reflection Questions

Subject: Introductory HCI & Computer Graphics

Slide 1: HCI Focus — Identifying Interaction & UI Elements

1. Search Bar

What it is: A prominent search input field located at the top-left of the interface where users can type any location name, address, or landmark to navigate directly to that place.

How it helps the user: The search bar dramatically reduces navigation time by eliminating the need to manually pan and zoom across the globe. It provides autocomplete suggestions and geocoding, enabling users to reach their desired location within seconds. This is a core usability feature that supports the user's primary goal of finding places efficiently.

2. Navigation Controls (Zoom & Compass)

What it is: A set of interactive controls located at the bottom-right, including zoom in/out buttons (+/-) and a compass widget that indicates the current orientation (North direction).

How it helps the user: These controls provide intuitive orbit navigation, allowing users to rotate, tilt, and zoom the 3D view with precision. The compass offers critical orientation feedback, ensuring users do not lose their spatial bearings while exploring complex 3D cityscapes. This makes spatial navigation effortless, especially for novice users unfamiliar with 3D interaction gestures.

3. Layers Panel

What it is: A collapsible side panel on the left that allows users to toggle various data layers on and off, such as Satellite imagery, 3D Buildings, Terrain, Roads, Borders, and more.

How it helps the user: The Layers Panel gives users direct control over the visual complexity of the interface based on their specific task. For example, a user studying topography can enable Terrain and disable 3D Buildings, while a tourist exploring a city might do the opposite. This adaptability reduces cognitive load and keeps the interface clean and relevant to the user's current objective.

Slide 2: CG Focus — Identifying Rendering & Visual Elements

1. 3D Mesh Geometry

What it is: The buildings, terrain, and landmarks in Google Earth are constructed from polygon meshes, composed of vertices, edges, and faces that mathematically define the 3D shape and volume of each object.

How it creates 3D space: Without 3D mesh geometry, the scene would appear as flat 2D imagery. The mesh gives objects structural depth, height, and form. When the camera moves around a building, the mesh ensures the object maintains its correct proportions and perspective from every angle, creating a convincing illusion of a three-dimensional world.

2. Texture Mapping & Materials

What it is: High-resolution satellite and aerial photographs are mapped (wrapped) onto the 3D mesh surfaces of terrain and buildings. This process is called texture mapping, where 2D images are applied as surface materials.

How it creates 3D space: Texture mapping adds realistic surface detail, such as grass, roads, water, brick, and glass, without requiring millions of additional polygons. The combination of photographic textures on geometric shapes tricks the human eye into perceiving depth, material properties, and real-world context, making the virtual environment immediately recognizable and believable.

3. Lighting & Shadow Rendering

What it is: Google Earth uses directional lighting to simulate the sun's position, which generates dynamic shadows cast by buildings, trees, and terrain features. The lighting model calculates how light interacts with surfaces.

How it creates 3D space: Shadows are one of the most powerful depth cues in visual perception. They help users understand the relative height of buildings, the curvature of terrain, and the spatial relationships between objects. Realistic lighting and shadows add dimensionality and time-of-day context to the scene, transforming a static model into a living, breathing digital twin of Earth.

Reflection Questions — Connecting Computer Graphics to User Experience

Question 1:

How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

High-quality Computer Graphics in Google Earth, such as realistic lighting, detailed texture mapping, and accurate 3D building meshes, greatly enhances the user's spatial understanding and navigation confidence. Realistic shadows and depth cues help users intuitively judge distances, building heights, and terrain elevation without needing explicit measurements. However, if the rendering quality exceeds the device's hardware capabilities, it can cause frame drops, lag, and stuttering during interaction, which significantly hinders the user experience and may frustrate users, particularly those with older devices or slower internet connections.

Question 2:

What is one simple HCI change you would make to improve the user interface?

One simple yet effective HCI improvement would be to add a 'Quick Tour' onboarding button for first-time users that demonstrates basic 3D navigation gestures, such as drag-to-rotate, scroll-to-zoom, and right-click-to-pan, using animated visual hints. This would significantly reduce the learning curve for novice users who may otherwise struggle to discover the full capabilities of Google Earth's 3D interface. By providing immediate, interactive guidance, users can become proficient within seconds, leading to improved usability, higher engagement, and a more satisfying overall experience.